

# Cross-Dataset Intrusion Detection

BASED ON: final.pptx (12 SLIDES) | Minor Project Synopsis | B.Tech CSE | USTM

## TEAM & MODULE ALLOCATION (from Slide 1 & Slide 10):

| Team Member & Roll No.                   | Assigned Timeline & Engineering Module         | Assigned Slides |
|------------------------------------------|------------------------------------------------|-----------------|
| Pynskhemlang Marbaniang (2023/BCSE/0005) | Weeks 1–4: Ingestion, Cleaning & Mapping       | Slides 1–3      |
| Tehjib Almas Junaid (2023/BCSE/0032)     | Weeks 2–6: Baseline Models & Cross-Transfer    | Slides 4–6      |
| Bamelaai Marbaniang (2023/BCSE/0045)     | Weeks 6–7: Feature Selection & CORAL Alignment | Slides 7–9      |
| Md Sabique Huda (2023/BCSE/0066)         | Weeks 4–9: Cross-Testing & Streamlit Dashboard | Slides 10–12    |

## SLIDE-BY-SLIDE SCRIPT (EXACTLY 12 SLIDES AS IN final.pptx)

### SLIDE 1: TITLE SLIDE

Speaker 1 / Any Member

[SLIDE CONTENT]:

- MINOR PROJECT SYNOPSIS on 'Cross-Dataset Intrusion Detection'
- Team: Pynskhemlang Marbaniang, Tehjib Almas Junaid, Bamelaai Marbaniang, Md Sabique Huda
- Guide: Mr. Abdul Kayum Ali, Assistant Professor | Dept of Engineering & Technology | USTM

[SPEECH]:

"Good morning, respected Chairperson, honourable guide Mr. Abdul Kayum Ali sir, and esteemed faculty members. We are here to present our Minor Project Synopsis titled 'Cross-Dataset Intrusion Detection.' Our team consists of Pynskhemlang Marbaniang, Tehjib Almas Junaid, Bamelaai Marbaniang, and Md Sabique Huda, from the Department of Computer Science and Engineering, University of Science & Technology, Meghalaya. Our research tackles a critical security problem: why machine learning models for cyber intrusion detection perform with near-100% accuracy in academic labs, yet fail catastrophically when deployed into real-world enterprise networks."

[ACTION: Next Slide]

## SLIDE 2: 1. INTRODUCTION — i. Background of the Study

Speaker 1: Pynskhemlang Marbaniang

[SLIDE CONTENT]:

- Network Intrusion Detection Systems (NIDS) as frontline perimeter defense
- Signature-based systems (Snort, Suricata, Zeek) — blind to zero-days
- ML/DL approaches (Decision Trees, Random Forests, SVM, XGBoost) — report 98%–99.5% accuracy on benchmarks (CICIDS2017, CSE-CIC-IDS2018, UNSW-NB15)

[SPEECH]:

*"Network Intrusion Detection Systems (NIDS) are the frontline defense inspecting network traffic in real time for malicious exploits, policy violations, and data exfiltration. Traditionally, networks relied on signature-based systems such as Snort, Suricata, and Zeek. While they have high precision on known attacks, they are fundamentally blind to zero-day exploits, evasive polymorphic malware, or slow reconnaissance. To overcome this, cybersecurity researchers shifted to Machine Learning and Deep Learning. Algorithms such as Decision Trees, Random Forests, Support Vector Machines, and XGBoost learn statistical flow patterns from network packets. In published academic papers, these models routinely report 98% to 99.5% accuracy on standard benchmark datasets like CICIDS2017 and UNSW-NB15. However, as we will see on the next slide, this near-perfect accuracy hides a dangerous operational illusion."*

[ACTION: Next Slide]

## SLIDE 3: ii. Problem Statement — The Same-Dataset Evaluation Illusion

Speaker 1: Pynskhemlang Marbaniang

[SLIDE CONTENT]:

- In the Lab: Models trained & tested on same dataset (80/20 splits, k-fold CV). Closed world. Memorize artifacts.
- In Deployment: Moved to a different network (cross-dataset), face covariate shift & concept drift.
- Collapse: Accuracy plummets from ~99% toward random guessing; MCC collapses to near-zero. (99% → near-random)

[SPEECH]:

*"Here is the problem statement at the heart of our project. In academic laboratory setups, models are trained and tested on splits from the EXACT SAME dataset — typically using an 80/20 partition or k-fold cross-validation. Because training and test data share the exact same network topology, background operating systems, packet generation tooling, and capture time window, the model is operating in a 'closed world.' Instead of learning generalizable attack signatures, the model memorizes simulator-specific artifacts. When that model is actually deployed into a different network — what we call the cross-dataset setting — it encounters severe covariate shift and concept drift. As a result, the accuracy plummets from 99% down toward random guessing (40% to 60%), and the Matthews Correlation Coefficient (MCC) collapses toward zero. A system that appeared foolproof in the lab completely fails in production."*

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**SLIDE 4: iii. Importance of the Present Work & iv. Scope of the Study****Speaker 2: Tehjib Almas Junaid****[SLIDE CONTENT]:**

- Why This Matters: Deployed NIDS that fail under drift create false confidence; closing gap is essential for trustworthy ML.
- Scope: 2 Datasets (CICIDS2017 & UNSW-NB15) | 5 Classifiers (DT, RF, LR, SVM, XGBoost) | Bidirectional Transfer | Active Mitigation Layer

**[SPEECH]:**

*"Why does this research matter? If a deployed intrusion detection system fails under network drift, it generates a false sense of security while leaving enterprise networks completely exposed. Diagnosing and actively closing this gap is essential if we want trustworthy, deployable cybersecurity systems — not just models that win academic leaderboard scores. Our scope is precisely defined across four pillars: 1. We evaluate the two premier global benchmark datasets: CICIDS2017 (with 80+ features) and UNSW-NB15 (with 49 features). 2. We benchmark five distinct classifier families: Decision Tree, Random Forest, Logistic Regression, SVM, and XGBoost. 3. We execute bidirectional transfer: testing both CICIDS to UNSW, and UNSW to CICIDS. 4. We engineer an active mitigation layer combining statistical feature selection, robust normalization, and CORAL covariance alignment."*

**[ACTION: Next Slide]****SLIDE 5: 2. Literature Review (Part 1)****Speaker 2: Tehjib Almas Junaid****[SLIDE CONTENT]:**

- 1. Verkerken et al. (2022) | Bidirectional cross-dataset evaluation, CICIDS2017 ↔ UNSW-NB15 | Accuracy plummets 28–42%; feature harmonization is mandatory.
- 2. Cantone, Marrocco & Bria (2024) | 4 classifiers pairwise across 4 CIC datasets | Intra MCC >99.6%; cross MCC collapses to near-zero (~40% best case).
- 3. Ahmim, Amangele & Moustafa (2024) | UNSW-NB15 & TON\_IoT with 7 shared features | Accuracy declines up to 35%; naive intersection fails.

**[SPEECH]:**

*"To place our work in context, we reviewed the key literature on cross-dataset NIDS: First, Verkerken et al. (2022) in the Journal of Network & Systems Management performed bidirectional cross-dataset evaluation between CICIDS2017 and UNSW-NB15. They proved that accuracy plummets by 28% to 42%, and they concluded that feature harmonization is mandatory. Second, Cantone et al. (2024) at IEEE CSR tested 4 classifiers across 4 datasets in the CIC family. They demonstrated that while intra-dataset MCC exceeds 99.6%, cross-dataset MCC collapses to near-zero (only ~40% in the best case). Third, Ahmim et al. (2024) evaluated UNSW-NB15 and TON\_IoT, but restricted their analysis to a naive intersection of only 7 features, suffering an accuracy decline of up to 35% and proving that crude feature intersection alone is insufficient."*

**[ACTION: Next Slide]****SLIDE 6: 2. Literature Review (Part 2)****Speaker 2: Tehjib Almas Junaid****[SLIDE CONTENT]:**

- 4. Bertoli, Pereira & Saotome (2022) | DoS models on CICIDS2017 tested on physical testbed | 25.63% degradation; models rely on non-transferable background features.
- 5. Sarhan, Layeghy, Moustafa & Portmann (2021) | NetFlow features (43 attributes) across 4 benchmarks | Heterogeneity is primary obstacle; standardization improves transfer.

**[SPEECH]:**

*"Continuing our literature review: Fourth, Bertoli et al. (2022) trained DoS models on CICIDS2017 and tested them on an independent physical testbed, logging a 25.63% average accuracy degradation because classifiers heavily rely on non-transferable background features. Fifth, Sarhan et al. (2021) explored standardized NetFlow-based feature extraction across four NIDS benchmarks, confirming that feature-set heterogeneity is the single biggest barrier to transferability."*

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## SLIDE 7: 3. Research Gap — Three Pillars

Speaker 3: Bamelaai Marbaniang

[SLIDE CONTENT]:

- Pillar I: Diagnosis vs. Remediation — Existing studies diagnose the collapse but treat it as an observational dead-end.
- Pillar II: Semantic Feature Ontology Mismatch — CICIDS (CICFlowMeter) vs UNSW (Argus/Zeek) incompatible schemas.
- Pillar III: No Bidirectional Robustness Benchmark — Most studies test only one direction; tree vs margin models unranked.

[SPEECH]:

*"A rigorous synthesis of these papers reveals three critical research gaps: Pillar I: 'Diagnosis vs. Remediation.' This directly answers why our project is unique. Previous authors — like Verkerken et al. (2022) and Cantone et al. (2024) — observed the collapse, documented it, and stopped. They treated it as an observational dead-end. Our project moves from observational diagnosis to ACTIVE ENGINEERING REMEDIATION. We don't just show that accuracy drops — we engineer methods to bring it back! Pillar II: 'Semantic Feature Ontology Mismatch.' CICIDS uses CICFlowMeter with 80+ features; UNSW uses Argus/Zeek with 49 features. No standardized semantic ontology currently bridges these two leading benchmark families. Pillar III: 'No Bidirectional Robustness Benchmark.' Prior works test unidirectional transfer and do not systematically benchmark which model architecture — tree-based like Random Forest and XGBoost versus margin-based like SVM — is intrinsically most resilient to domain shift."*

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## SLIDE 8: 4. Feasibility Study

Speaker 3: Bamelaai Marbaniang

[SLIDE CONTENT]:

- Technical: Public datasets, mature Scikit-Learn/XGBoost, <6 GB RAM via stratified subsampling
- Economic: 100% open-source toolchain, free datasets, Colab T4 — net budget ■0.00
- Operational: 4 modules map to 4 members, GitHub workflow, Streamlit dashboard
- Schedule & Risk: 8–10 weeks; imbalance & memory mitigated via SMOTE-NC, robust scaling

[SPEECH]:

*"We conducted a comprehensive feasibility study across four dimensions: 1. Technical: Both datasets are publicly hosted by university portals without licensing fees. All five algorithms are natively implemented in Scikit-Learn and XGBoost. Memory footprint is kept under 6 GB RAM using stratified subsampling (200k–500k records) and streaming chunk ingestion. 2. Economic: The entire software stack (Python, Pandas, Scikit-Learn, Streamlit) is open-source. Free-tier Google Colab T4 GPUs provide auxiliary compute. Total project expenditure is exactly ■0.00. 3. Operational: Decomposed into 4 modular engineering tasks mapped directly to our 4 team members, coordinated via Git and GitHub. 4. Schedule & Risk: Planned across an 8-to-10 week semester timeline. Severe class imbalance is resolved through stratified undersampling and SMOTE-NC; feature scale disparities are handled via Robust and Quantile scaling."*

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**SLIDE 9: 5. Objectives****Speaker 3: Bamelaai Marbaniang****[SLIDE CONTENT]:**

- 1. Analyze and mathematically formulate single-dataset failure (covariate shift, concept drift, artifacts).
- 2. Design bidirectional cross-dataset testbed harmonizing CICIDS2017 & UNSW-NB15 into shared feature space.
- 3. Benchmark 5 classifiers (DT, RF, LR, SVM, XGBoost) on same & cross-dataset regimes using Acc, F1, MCC, FAR.
- 4. Apply feature selection & domain alignment (CORAL) to close gap; deploy Streamlit dashboard.

**[SPEECH]:**

*"Our project is guided by four specific, actionable objectives: Objective 1: Analyze and mathematically formulate why single-dataset cross-validation fails to predict real-world performance, identifying the exact statistical mechanisms of covariate shift and artifact overfitting. Objective 2: Design and construct a standardized bidirectional testbed that reconciles heterogeneous features from CICIDS2017 and UNSW-NB15 into a unified semantic space. Objective 3: Implement, benchmark, and evaluate five representative ML classifiers across both intra-dataset baselines and cross-dataset transfer regimes using comprehensive metrics: Accuracy, Precision, Recall, Macro-F1, MCC, and False Alarm Rate. Objective 4: Formulate and empirically validate feature selection and CORAL domain-alignment techniques to close the generalization deficit, deploying an interactive Streamlit dashboard for real-time demonstration."*

**[ACTION: Next Slide]****SLIDE 10: 6. Methodology / Planning of Work****Speaker 4: Md Sabique Huda****[SLIDE CONTENT]:**

- 7-Step Pipeline: Acquisition → Mapping → Baselines → Cross-Transfer → Alignment → Retrain (GRR) → Dashboard
- Symmetric directions: CICIDS2017 → UNSW-NB15 and UNSW-NB15 → CICIDS2017
- Timeline Table: W1–2 (Pynskhemlang), W2–4 (Pynskhemlang, Tehjib), W4–6 (Tehjib, Sabique), W6–7 (Bamelaai), W7–9 (All)

**[SPEECH]:**

*"Our research methodology follows an end-to-end 7-step pipeline: Step 1: Ingest official CSV records, cap infinite flow rates, and clean missing values. Step 2: Build the semantic mapping dictionary linking equivalent features (e.g., CICIDS 'Flow Duration' ↔ UNSW 'dur'; 'Flow Packets/s' ↔ 'rate'). Step 3: Establish the same-dataset performance ceiling (expected 97%–99%). Step 4: Execute zero-shot bidirectional cross-dataset transfer (CICIDS → UNSW and UNSW → CICIDS), exposing the performance drop. Step 5: Apply Mutual Information feature selection, Quantile transformation, and Correlation Alignment (CORAL) to align feature covariance matrices without target labels. Step 6: Retrain models on aligned features and calculate the Generalization Recovery Ratio (GRR) to measure how much lost performance was recovered. Step 7: Package the pipeline into an interactive Streamlit web dashboard. As shown in our 9-week timeline, each phase is assigned to specific team members to ensure smooth and timely execution."*

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## SLIDE 11: 7. Expected Outcome

Speaker 4: Md Sabique Huda

[SLIDE CONTENT]:

- 1. Bidirectional Evaluation Harness: Automated Python pipeline for cross-dataset benchmarking in both directions.
- 2. Empirical Generalization Map: Quantified collapse across 5 ML models.
- 3. Feature Alignment Toolkit: Reconciling CICFlowMeter & Argus/Zeek with CORAL.
- 4. Robustness Ranking: Empirical ranking of architectures under covariate shift.
- 5. Streamlit Dashboard: Production-ready web app for real-time simulation.

[SPEECH]:

*"Upon completion, our project will deliver five tangible engineering and scientific outcomes: Deliverable 1: A modular, automated Python testbed that standardizes cross-dataset benchmarking in both directions. Deliverable 2: A comprehensive empirical map quantifying the performance deficit across 5 machine learning models using MCC, Macro-F1, and False Alarm Rates. Deliverable 3: An open-source semantic feature alignment toolkit that bridges CICFlowMeter and Argus/Zeek schemas, integrated with CORAL domain adaptation. Deliverable 4: A clear empirical ranking showing which machine learning architectures (tree ensembles vs. linear models vs. SVM) possess the greatest intrinsic resilience under network domain shift. Deliverable 5: A live, interactive Streamlit Security Operations Dashboard where analysts can select source/target datasets, toggle domain alignment on and off, and inspect confusion matrices and ROC curves in real time."*

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## SLIDE 12: 8. References

Speaker 4: Md Sabique Huda

[SLIDE CONTENT]:

- References: Ahmim (2024), Bertoli (2022), Cantone (2024), Sarhan (2021), Verkerken (2022)
- Department: Computer Science & Engineering, USTM
- Session: 2026–2027

[SPEECH]:

*"Our work builds upon the foundational research listed here, strictly formatted in accordance with the APA 7th Edition citation standard: Verkerken et al. (2022) in IEEE Transactions on Network and Service Management, Cantone et al. (2024) in IEEE CSR, Bertoli et al. (2022) in Software Quality Journal, Ahmim et al. (2024) in IEEE Access, and Sarhan et al. (2021) in Mobile Networks & Applications. In conclusion, our project moves machine learning intrusion detection beyond the illusory comfort of laboratory benchmarks toward deployable, domain-resilient security. On behalf of Pynskhemlang, Tehjib, Bamelalai, and myself, thank you very much for your time and guidance. We are now happy to answer your questions."*

[END OF PRESENTATION SPEECH]

## EXAMINER DEFENSE CHEATSHEET (SPECIFIC TO final.pptx)

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**Q1: "In Slide 5, you cite Verkerken et al. (2022) who already tested CICIDS2017 ↔ UNSW-NB15. How is your work different or new?"**

**A1:** "Sir/Madam, Verkerken et al. (2022) conducted an observational study. They observed that accuracy drops by 28–42% and noted that 'feature harmonization is mandatory', but they did NOT implement an active solution. They left it as an open question. Our project represents Research Gap I: Diagnosis vs. Remediation (Slide 7). We actively construct the semantic ontology, apply CORAL covariance alignment, evaluate the Generalization Recovery Ratio (GRR), and deploy an interactive Streamlit tool. Verkerken diagnosed the problem; we engineer the remedy."

**Q2: "In Slide 4, why did you choose CICIDS2017 and UNSW-NB15 specifically?"**

**A2:** "Because they represent the two most influential but structurally incompatible benchmark families globally. CICIDS was captured on a 1 Gbps network using CICFlowMeter (80+ features), whereas UNSW was captured on a 10 Gbps testbed using Argus/Zeek (49 features). Testing transfer between them represents the most authentic simulation of deploying a model into an unseen network."

**Q3: "In Slide 10, how does CORAL alignment work without target labels?"**

**A3:** "CORAL is unsupervised. It calculates the covariance matrix of the source traffic and the target traffic. It whitens the source features and recolors them using the target covariance. Because it only needs traffic distributions and NOT labels, it is 100% practical for real-world deployment where incoming target network traffic is unlabeled."